

A Complex-Valued Neural Network for Motor Imagery Decoding in EEG



G. A. MENDOZA FRANCO¹, X. SUN¹, G. AGARWAL²

¹ Pitzer–Scripps Dept. of Natural Sci., Pitzer College, Claremont, CA

² Dept. of Natural Sciences, Pitzer and Scripps Colleges, Claremont, CA



Introduction

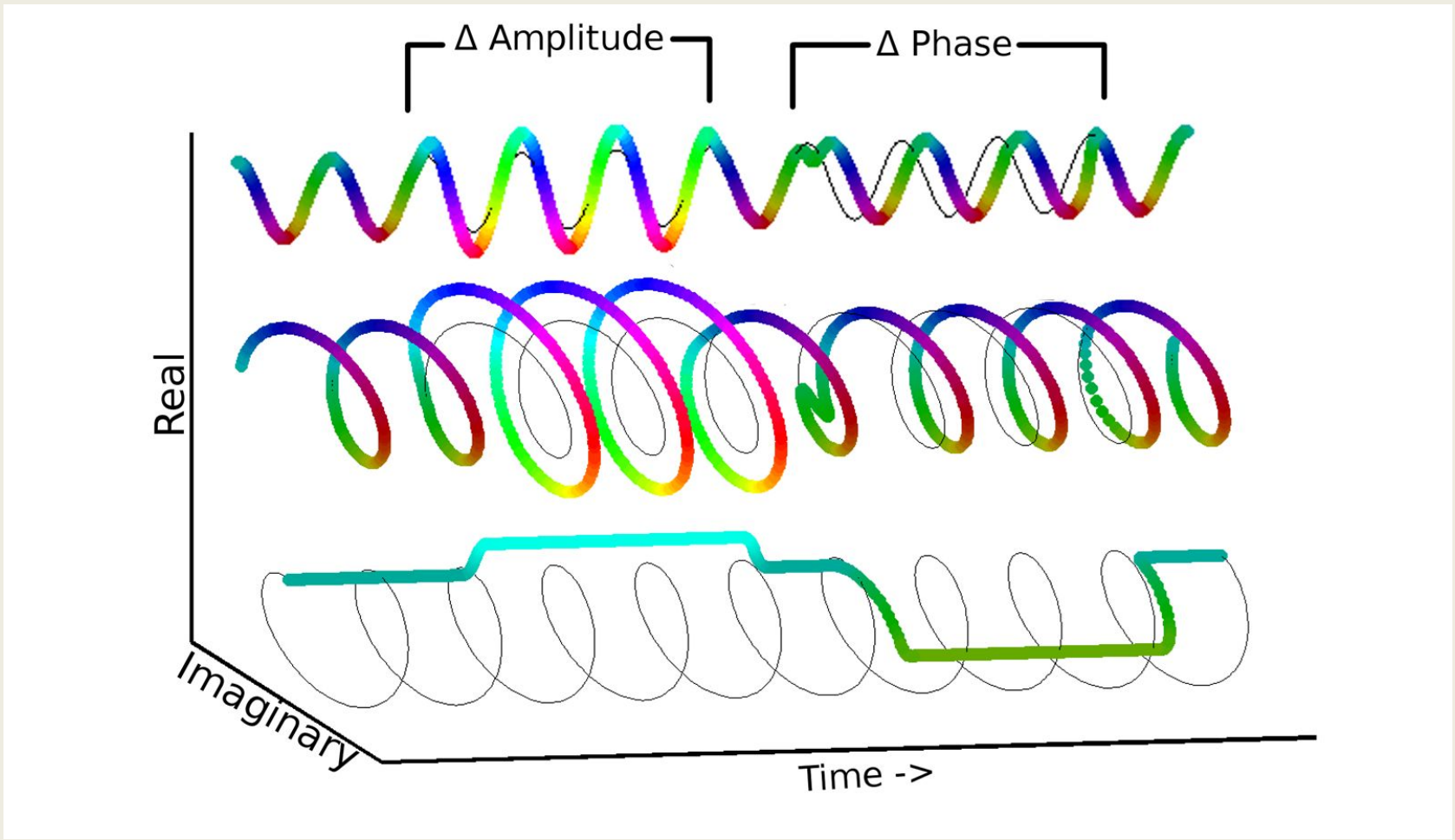
- Oscillatory brain signals, such as local field potentials (LFPs) and electroencephalography (EEG), reflect behavior through patterns of phase and amplitude across electrodes. **TIMBRE**, a complex-valued neural network, previously showed superior position decoding of rat hippocampal 8Hz LFPs compared to Common Spatial Patterns (CSP).^{1,3}
- Here, we evaluate the extent to which **TIMBRE's** superior performance generalizes to human EEG data relative to classical methods.
- Understanding this helps identify when complex-valued neural decoders are useful, clarifies the value of cCSP and TIMBRE for EEG, and guides the design of oscillation-based brain-computer interfaces.

Dataset

- BCI Competition III Dataset IVa²
- 118-channel EEG from 5 healthy participants.
- Cued motor imagery trials of right hand vs. right foot movements (binary classification)
- Signals were recorded with dense scalp coverage and resampled to 100 Hz.
- We focused on the 3 participants with the largest number of labeled trials.

BCI Competition III Dataset IVa subjects with labeled training trials (#tr) and unlabeled test trials (#te). Subjects selected for analysis are outlined in red.	#tr	#te
aa	168	112
al	224	56
av	84	196
aw	56	224
ay	28	252

Complex Values



Complex-valued representations allow for capturing spatio-temporal relationships within waves.

Relevance:

- Complex-valued representations preserve both amplitude and phase of EEG signals via the Hilbert transform, allowing models to capture timing relationships across channels.
- This approach extends prior LFP work in the hippocampus, where incorporating phase improved the decoding of theta-band oscillations.¹

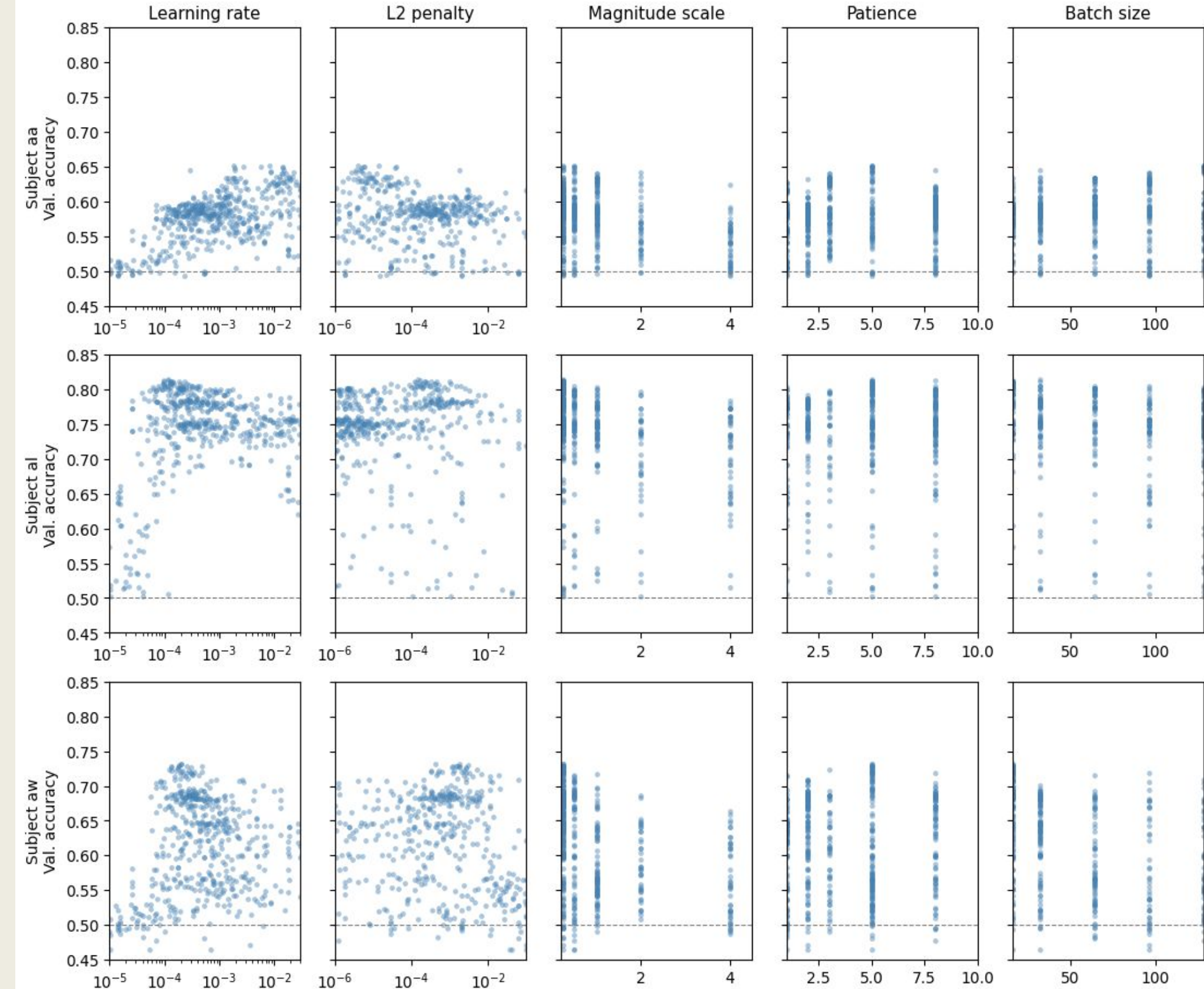
Models

CSP is a classification method that learns spatial filters which maximize variance for one class while minimizing it for another.³ We introduce two variables for model comparison:

- Real-valued CSP (**rCSP**) uses band-pass filtered EEG as input and serves as our standard baseline.
- Complex-valued CSP (**cCSP**) uses Hilbert-transformed complex EEG. For each class, we compute complex covariance matrices and solve a generalized eigenvalue problem to obtain complex spatial filters.

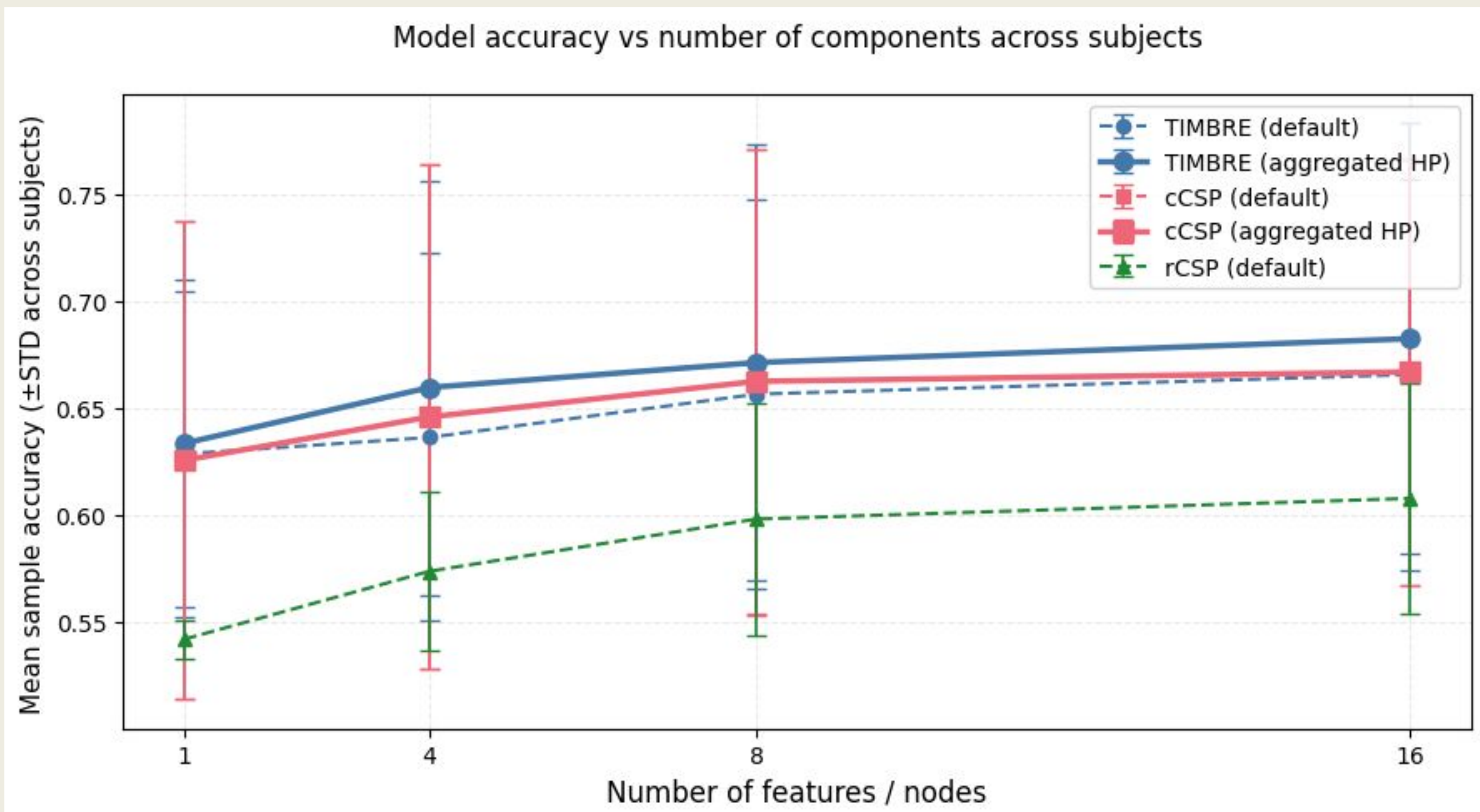
TIMBRE is a complex-valued neural network that learns spatial and phase patterns from multichannel oscillations that carry information about a variable of interest. It was developed and validated on multi-electrode LFP recorded from rat hippocampus, where it outperformed CSP.

Hyperparameter Optimization



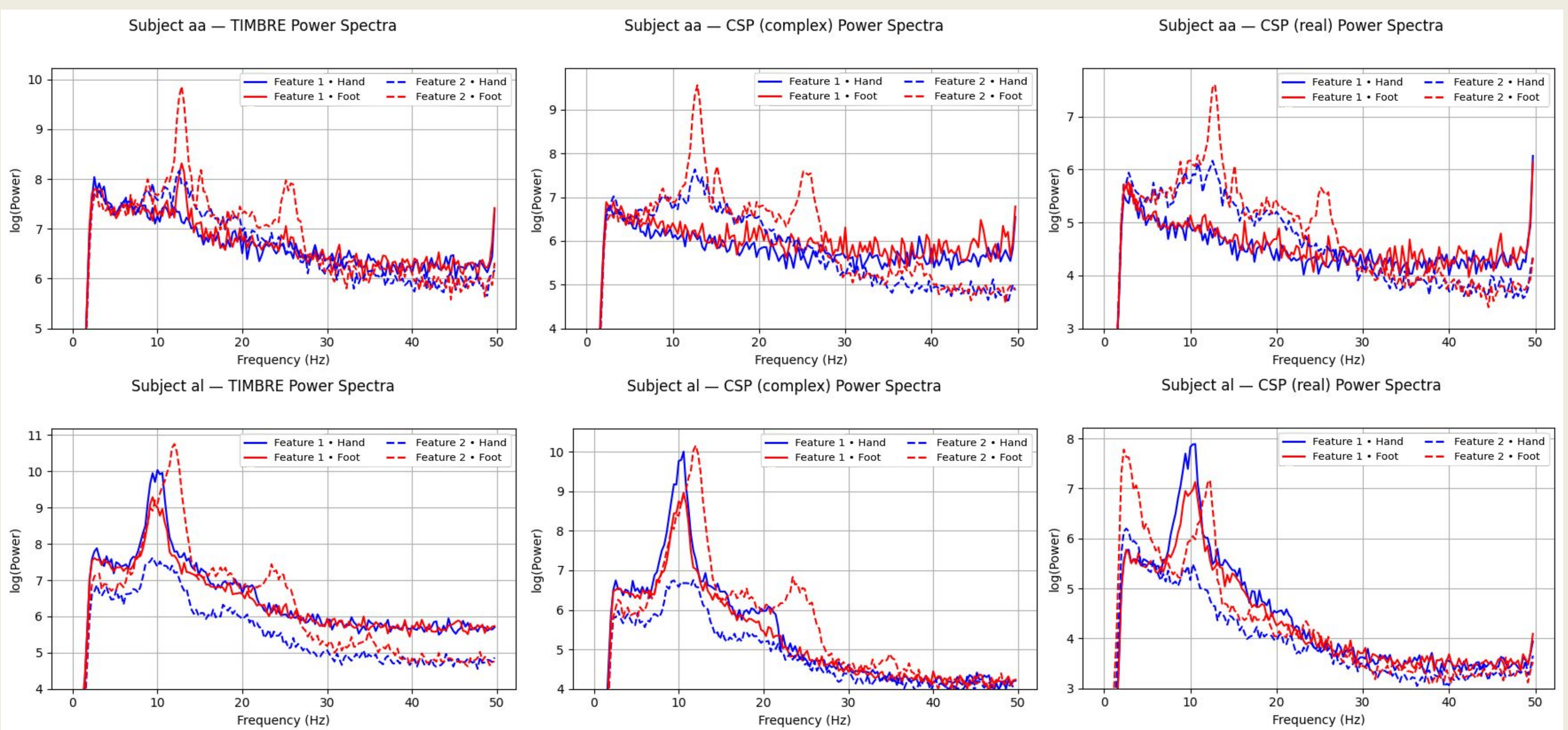
TIMBRE hyperparameter sensitivity across subjects. **Learning rate and L2 penalty dominate sensitivity; subject-specific accuracy ceilings indicate data-driven, not search-driven, performance limits.**

Model Accuracy across subjects



Complex-valued methods achieve higher decoding accuracy in motor-imagery classification. Mean sample-level accuracy (5-fold cross-validation, averaged across subjects) for default and aggregated hyperparameter configurations TIMBRE, cCSP, and rCSP.

Power spectra



Complex models capture motor-related rhythms at 9–12 Hz and 22–25 Hz, though component selectivity varies across subjects. Solid/dashed lines represent separate components (Class 0 (hand) = red, Class 1 (foot) = blue). Complementary activations between line types reflect class selectivity; inconsistencies across subjects may indicate variable motor imagery engagement.

Discussion

- Complex-valued methods outperform rCSP across all subjects and component counts, confirming phase information carries class-discriminative structure beyond amplitude alone
- After optimization, TIMBRE matches or modestly exceeds cCSP; gains are subject-dependent and overlap within fold-level standard deviation with minimal transferrability
- Spectral analysis of learned projections reveals statistically significant ($p < 0.05$, FDR-corrected) **class-dependent power differences** concentrated in **mu (9–12 Hz)** and **beta (20–25 Hz)** bands

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References

- Agarwal G, Lustig B, Akera S, Pastalkova E, Lee AK, Sommer FT. News without the buzz: reading out weak theta rhythms in the hippocampus. bioRxiv [Accepted in Principle to *Nature Communications* 2025].;
- B. Blankertz, et al. The BCI competition 2003: Progress and perspectives in detection and discrimination of EEG single trials. IEEE Trans. Biomed. Eng., 2004.;
- B. Blankertz, et al. "Optimizing Spatial filters for Robust EEG Single-Trial Analysis," in IEEE Signal Processing Magazine, 2008.